

$$* I_{S,E}, \quad L = \int d\phi \int dr r I(r)$$

• Total Luminosity $R \rightarrow \infty$

$$* \text{Spirals: } \underline{L_{tot}^s} = 2\pi R_h^2 I_0 \leftarrow b=1$$

$$* \text{Ellipticals: } \underline{L_{tot}^e} = 7.22 \pi I_e R_e^2 \leftarrow \left(\frac{1}{n} e^{-b} \frac{\Gamma(1)}{\Gamma(n)} \right)$$

$$* \left[\frac{M}{L} \right]: \text{Mass relationship, Spirals } M = \frac{V^2}{R}$$

$n=2$: spiral

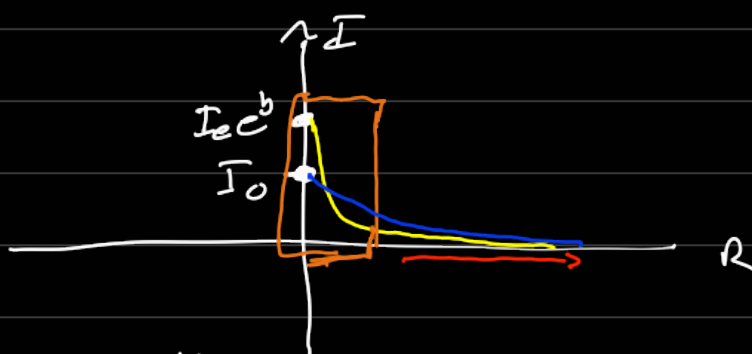
$n=4$: ell.

$$I_s(R) = I_0 e^{-R/R_h}$$

$$I_e(R) = I_e e^{-b(R/R_h)^{1/4}} e^{+b}$$

$$f(x) = e^{-R/R_h}$$

$$I_0 > I_e e^b$$



$$I_e = I_e e^b e^{-b(R/R_h)^{1/4}}$$

$$I(0) = \underline{I_e e^b}$$

$$* I_s(R) = I_0 e^{-R/R_h}$$

$$I_s'(R) = -I_0 \frac{1}{R_h} e^{-R/R_h} = -\frac{1}{R_h} I(R)$$

$$I_e'(R) = -b \frac{1}{4} (R/R_h)^{1/4-1} e^{-b(R/R_h)^{1/4}} I_e$$

$$= -\frac{b}{4} \left(\frac{R}{R_h} \right)^{-3/4} I_e = -\frac{b}{4} \left(\frac{R_h}{R} \right)^{3/4} I_e$$

* CAS: *

(e) late type stars $\rightarrow$ Red $z >$

$\Rightarrow$ Ell.: red * "Early-type"

Spiral: blue *

* Towards [M/L]

$\rightarrow$ Mass function: $2u+1$

Elliptical: $K = \frac{1}{2} M_{gal} V_{rms}^2$

$$V_{rms}^2 = \langle V_x^2 + V_y^2 + V_z^2 \rangle = 3 \sigma_r^2$$



$$* K = \frac{1}{2} M_{gal} 3 \sigma_r^2$$

$$U = - \alpha \frac{G M_{gal}^2}{R}$$

$\alpha \approx 1$, $\frac{3}{5} \rightarrow$ Assumed a sphere

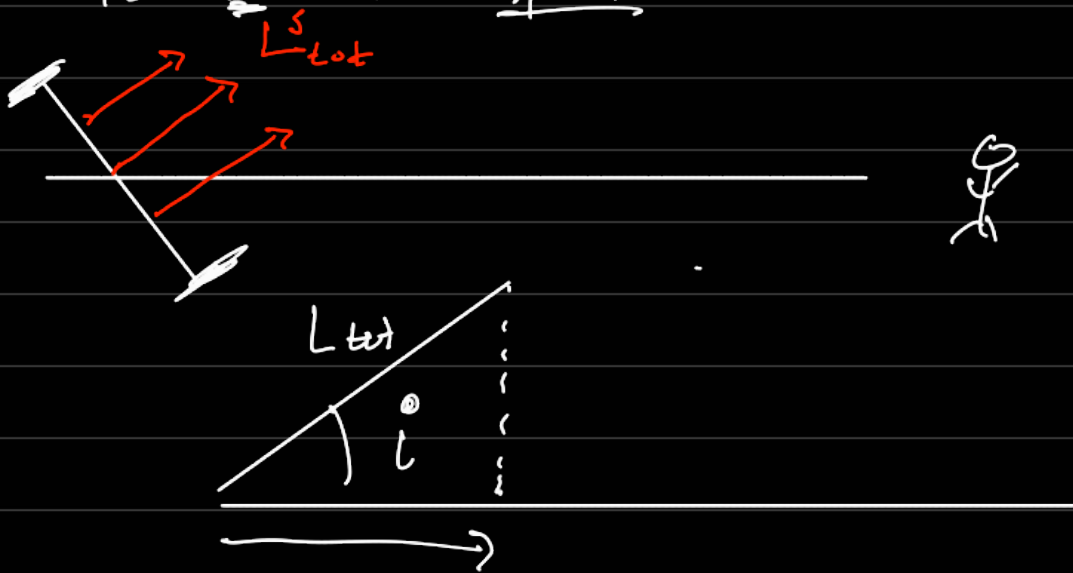
cd: $2K + U = 0 \Rightarrow 3 \sigma_r^2 M_{gal} = \alpha \frac{G M_{gal}^2}{R}$

Solve for M_{gal}

$$* M_{gal} = \frac{3 \sigma_r^2 R}{\alpha} *$$

First $[M/L]$

Going back to $\frac{L}{L_{tot}}$ of spiral:

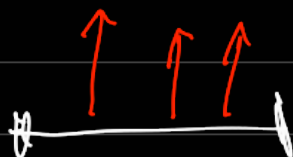


$$\cos(i) = \frac{L_{obs}}{L_{tot}} \Rightarrow L_{obs} = L_{tot} \cos(i)$$

or
*

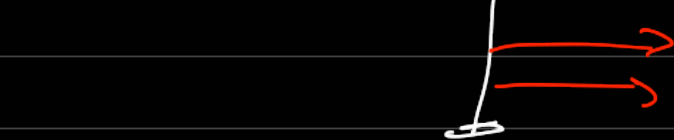
$$L_{tot} = \frac{L_{obs}}{\cos(i)} *$$

$$i = 90$$



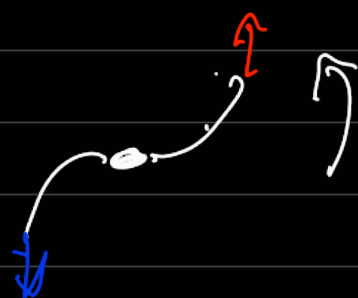
$$i = 0:$$



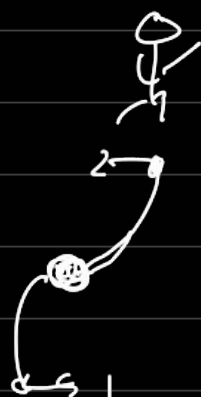


Velocity of star:

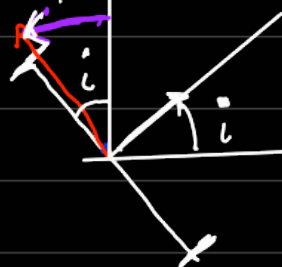
$$i = 90$$



$$i = 0$$



* General i :



$$V_{obs} = \sin(i)$$

V_{true}

$$\Rightarrow V_{obs} = \sin(i) V_{true}$$

Learn:

- $i = 90$ bad
- $i = 0$ good

Vel:

$i = 0$: bad

$i = 90$: good

Tully-Fisher: $L \propto V^4$, (Spirals)

$$\text{Vir thm: } 2K + U = 0$$

$$M_{gal} V^2 = 2 \frac{G M_{gal}^2}{R}$$

$$K = \frac{1}{2} M_{gal} V^2$$

$$V^2 = \frac{2 G M_{gal}}{R}$$

$$\frac{V^2}{L} = \frac{2 G \mu}{R} = \frac{M}{L}$$

$\mu = \frac{M}{L}$, divide by L :

L can be given by $L_{tot} = 2\pi R^2 I$

$$R \approx L_{tot}^{1/2}$$

$$\hat{L} = 2 G (2\pi I_0)^{1/2}$$

$$V^2 = \frac{\hat{L} \mu}{R^{1/2}} \Rightarrow \text{square}$$

$$V^2 = \frac{\tilde{c} \mu L}{L^{1/2}} = \tilde{c} \mu L^{1/2}$$

$$V^4 = (\tilde{c})^2 \mu^2 L \Rightarrow \boxed{\frac{V^4}{\mu^2} \tilde{c}^{-2} = L}$$

$$L \propto \frac{V^4}{\mu^2} \quad (\text{B. 2(b) PE})$$

Tully - Fisher

(i) Viral: $RK + U = 0$

$$V^2 = \frac{\alpha G M_{gal}}{R}$$

(ii) Divide by L: $\frac{V^2}{L} = \frac{\alpha G \left(\frac{M_{gal}}{L} \right)}{\mu}$

$$\frac{V^2}{L} = \frac{\alpha G \mu}{R}$$

(iii) $L_{tot} = 2\pi R^2 I_0 \Rightarrow R = \left(L_{tot} / I_0 \right)^{1/2} \frac{I}{(2\pi)^{1/2}}$

$$\frac{V^2}{L_{tot}} = \frac{\alpha G \mu (2\pi)^{1/2}}{(L_{tot} / I_0)^{1/2}} = \frac{\alpha G \mu (2\pi)^{1/2} I_0^{1/2}}{L_{tot}^{1/2}}$$

$$V^2 = \beta \frac{\mu I_0^{1/2} L_{tot}}{L_{tot}^{1/2}}$$

$$\boxed{\beta = \alpha G (2\pi)^{1/2}}$$

$$V^2 = \beta \mu I_0^{1/2} L_{tot}^{1/2}$$

Square:

$$V^4 = \beta^2 \mu^2 I_0 L_{tot}$$

$$\boxed{L_{tot} = \frac{V_{max}^4}{\beta^2 \mu^2 I_0}}$$

* Gal.: $\left[\frac{M}{L} \right]$



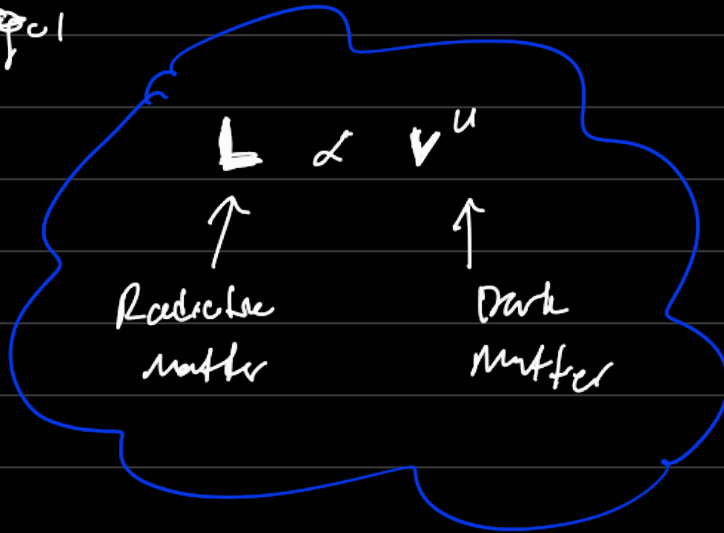
L_{tot} : Radiative matter

→ Relationship is a tool to gauge the relative amounts of radiative and dark matter

$$\left[\frac{M}{L}\right]_{sun} \sim 1-2 \quad (sun)$$

$$\left[\frac{M}{L}\right]_{gal} \sim \geq 20$$

μ :



$$(p_{dm})(R_0) \approx \frac{0.006 M_\odot}{\text{pc}^3}$$

